WorksheetDownload.com

10 CONCEPTUAL Q&A OPERATING SYSTEMS · PRACTICE WORKSHEET

CONCEPTUAL SERIES
Ubuntu

Name: ____________ Cohort/Batch: ____________ Date: ____________ Score: ____

UBUNTU PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Operating Systems

TOTAL: 10 QUESTIONS

Q1 What is Ubuntu and what problem in Operating Systems does it solve? *Threat Model*

Q2 What are the core components or concepts in Ubuntu? *Architecture*

Q3 How does Ubuntu compare to its main alternatives? *Configuration*

Q4 How do you get started with Ubuntu for a new project? *Hardening*

Q5 What configuration options most affect Ubuntu's behavior in production? *SSO / IdP*

Q6 How does Ubuntu handle reliability and high availability? *Observability*

Q7 What observability does Ubuntu provide out of the box? *Lifecycle*

Q8 What are common performance bottlenecks in Ubuntu? *Compliance*

Q9 What security best practices apply when running Ubuntu? *Testing*

Q10 What mistakes do teams commonly make when adopting Ubuntu? *Tuning*

ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Ubuntu is a tool or platform that

Mitigation

Ubuntu is a tool or platform that automates or simplifies a specific concern in the Operating Systems domain, reducing manual effort and operational complexity for engineering teams.

A2 Ubuntu has key building blocks that work

Primitives

Ubuntu has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A3 Ubuntu trades off simplicity against feature richness

Hardening

Ubuntu trades off simplicity against feature richness differently than its alternatives. Choose Ubuntu when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A4 Install Ubuntu via its package manager or

Gotchas

Install Ubuntu via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A5 Production-critical settings typically include concurrency

Federation

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A6 Ubuntu provides HA through leader election, replication,

Observability

Ubuntu provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A7 Ubuntu exposes metrics (Prometheus endpoint or built-in

Automation

Ubuntu exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A8 Performance bottlenecks typically stem from misconfigured

Performance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A9 Run Ubuntu with the principle of least

Assessment

Run Ubuntu with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A10 Common pitfalls include skipping the documentation and

Optimization

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.